

ChE354 Transport Process

Spring 2022

Transport process is an extension of ChE 319. We apply the concepts learned in momentum transfer and in heat transfer to solve practical chemical engineering problems.

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Office hours:

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TA:

Lectures:

MWF 12:00 – 1:00 PM

Please be punctual. Attendance is not compulsory, but is strongly recommended.

Bring your Calculator to class.

Recommended Calculator (TI-Nspire CX CAS).

Should be able to solve system of linear and non-linear algebraic equations.

Should be able to do numerical integration.

Recitation:

W 1:00PM- 2:00PM CPE 2.216.

Bring your laptop to recitation. We will use polymath to solve problems.

Optional Books:

Chemical Process Equipment Design, Turton, R, Shaeiwitz, J, Prentice Hall, 2017.

Fundamentals of Heat and Mass transfer, 6th Edition, Frank P. Incropera, David P. DeWitt, Theodore L. Bergman, John Wiley 2007.

Homework: Weekly homework will be posted on Canvas on Wednesday and are due on next Wednesday at the beginning of the class. Late homework will not be accepted. Homework must be neat and presented in the standard ChE format.

Quizzes: Quizzes will be given once a week during recitation.

Exams: There will be two midterm exams (2 hours and 30 min) and a final exam (3 hours). Exams are closed book and closed notes.

Exam – I	Feb 17 th	(7:30 – 10:00 PM) CPE. 2.218
Exam – II	April 7 th	(7:30 – 10:00 PM) CPE. 2.218
Final Exam	May 11 th	(2:00 – 5:00 PM)

Final Grade

Homework	10 %
Weekly Quizzes + mini Projects	20 %
Midterm Exams	40 %
Final Exam	30 %

Final letter grade will be given based on Class average. The class average for this course is B-. There are two requirements for a passing grade C-: You should have an Exam grade of 60 points and have a course average that will be determined from your class average and standard deviation.

Absence: No excuse for missed exams. If you miss an exam due to health or family reason, your final exam score will be used for the missing exam.

Topics to be covered in Transport Processes

- 1) Review of heat transfer mechanisms and steady-state one dimensional heat conduction
 - Fourier's Law, Convection, Radiation
 - Heat conduction in composite walls (Rectangular, Cylindrical, Spherical)
 - Overall heat transfer coefficient
 - Optimum insulation thickness
 - Heat conduction in extended surfaces (Fins etc)
- 2) Two-dimensional, steady-state conduction
 - Conduction Shape Factor
 - Numerical Solution (finite-difference equations)
- 3) Transient heat conduction
 - Lumped Capacitance Method
 - Semi-Infinite geometry (Error function solution, Numerical Solutions)
- 4) Convective heat transfer
 - Boundary Layer analysis
 - External flow (flow around immersed objects)
 - Internal flow (flow through conduits)
 - Natural convection
 - Boiling and Condensation
- 5) Radiation
 - Blackbody radiation
 - Radiation from real surfaces
 - Radiation exchange between surfaces (View Factor calculation)
- 6) Analysis and Design of Heat exchangers
 - Double-Pipe exchangers, Shell and Tube, Reboilers, Evaporators

- LMTD, Effectiveness-NTU calculations
 - Network of heat exchangers
- 7) Dimensional Analysis and similitude
- Buckingham's Pi method
 - Volumetric flowrate through a pipe
 - Heat transfer inside a pipe
- 8) Fluid Statics and Dynamics
- Manometers, variation of pressure due to elevation and rotation
 - Conservation of mass, tank filling
 - Conservation momentum, calculating force on bends and reducers.
 - Application of Bernoulli's equation and mechanical energy balance.
 - Flow meters: Pitot tube, orifice and venturi meter, rotameter
 - Fluid Friction in pipes, piping and pumping problems, flow in noncircular ducts
- 9) Pumps and Compressors
- Pump selection, pumps in series and parallel
 - Pump curves and System Curves
 - NPSH, application of Affinity Laws
 - Multi-stage compressors, optimum inter-stage pressure
- 10) Flow around immersed objects
- Drag force on sphere, cylinders, and calculation of drag coefficient.
 - Stokes' and Newton's region, application to gravity setting tanks, Centrifugal Separation, Cyclone and hydro cyclone separator
- 11) Packed beds and Fluidization
- Characterization of particles: sphericity, particle diameter, size distribution
 - Flow through porous medium, Darcy's law, permeability factor.
 - Frictional loss for packed bed: Kozeny-Carman Equation, Burke-Plummer equation and Ergun Equation.
 - Application of MEB for packed beds.
 - Fluidization state calculation of minimum fluidization velocity
 - Frictional loss and Pumping requirement for fluidized bed.
- 12) Non-Newtonian fluids
- Power Law fluids, Bingham Plastics fluid, Slurry
 - Pipe flow and pumping power calculation, external flows
- 13) Compressible fluids
- Ideal gas, isothermal, adiabatic and isentropic flow. speed of sound
 - Pipe flow: Isothermal flow, Adiabatic flow and Choked Flow
 - Flow through converging-diverging nozzle
 - Normal shock